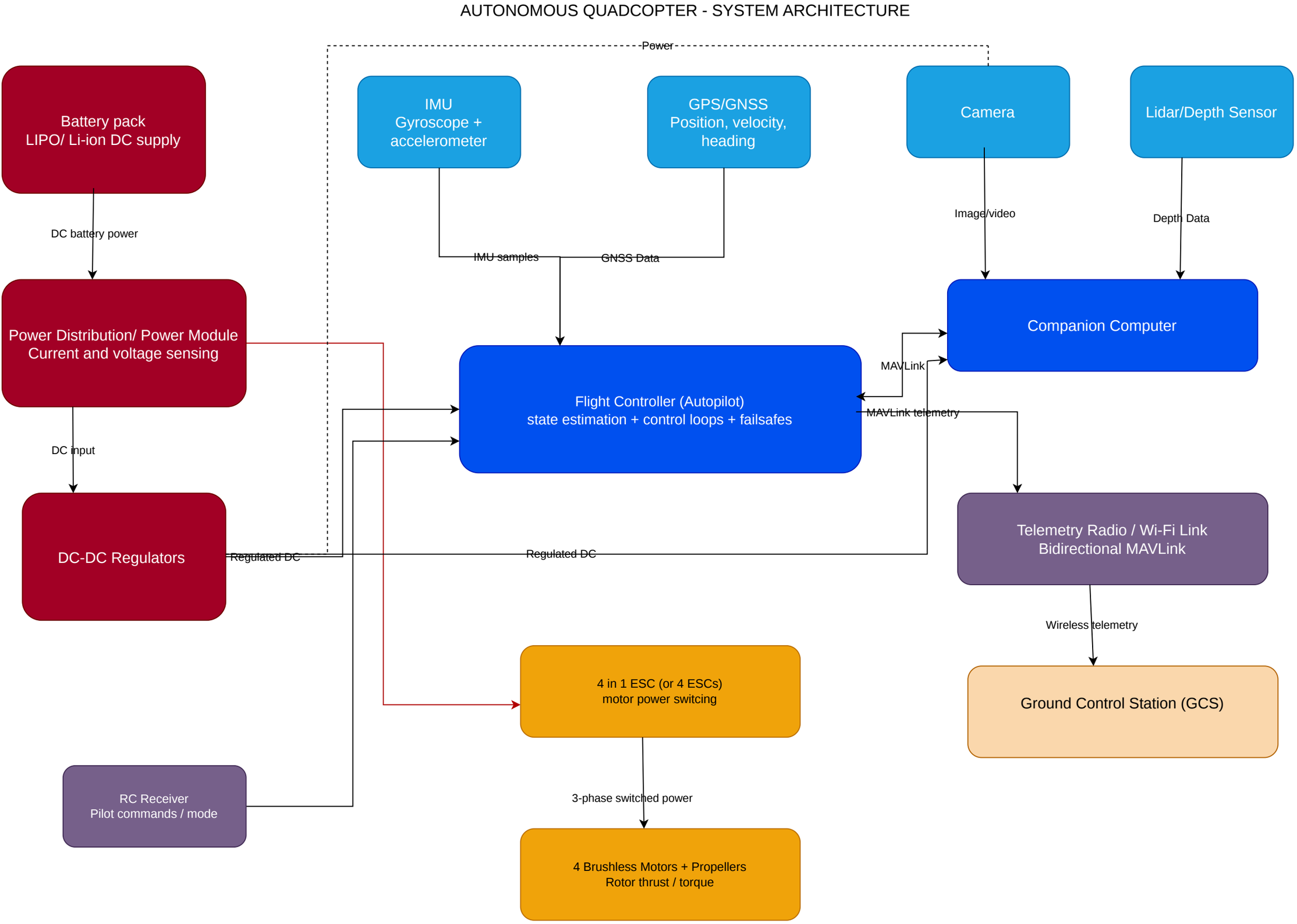
AUTONOMOUS QUADCOPTER - SYSTEM ARCHITECTURE
Power
Battery pack
LIPO/ Li-ion DC supply
DC battery power
Power Distribution/ Power Module
Current and voltage sensing
DC input
DC-DC Regulators
Regulated DC
IMU
Gyroscope +
accelerometer
GPS/GNSS
Position, velocity,
heading
IMU samples
GNSS Data
Flight Controller (Autopilot)
state estimation + control loops + failsafes
Regulated DC
Camera
Lidar/Depth Sensor
Image/video
Depth Data
Companion Computer
MAVLink
MAVLink telemetry
Telemetry Radio / Wi-Fi Link
Bidirectional MAVLink
Wireless telemetry
Ground Control Station (GCS)
4 in 1 ESC (or 4 ESCs)
motor power switcing
3-phase switched power
4 Brushless Motors + Propellers
Rotor thrust / torque
RC Receiver
Pilot commands / mode